

Luckie Musngi

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EDUCATION

University of Arizona (Graduated May 2026)

Tucson, AZ

- **Bachelor of Science** in Computer Science, **Minor** in Mathematics
- **Relevant Coursework:** Operating Systems, Database Design, Computer Networks, Web Programming

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, C/C++, SQL, HTML/CSS, MIPS Assembly

Frameworks & Tools: React.js, MongoDB, Unity, NumPy, Pandas, Matplotlib, Scikit-learn, Git/GitHub

Hardware & Systems: Embedded Systems, CAD (Fusion 360, OpenSCAD), Arduino

EXPERIENCE

University of Arizona

Tucson, AZ

Undergraduate Researcher, Bio Robotics & Scientific Instrumentation

Jan 2026 - May 2026

- Designed and tested robotic systems for automating biology and chemistry experiments.
- Evaluated and reported system reliability and repeatability through iterative experimental testing.
- Integrated motors, actuators, and sensors into automated experimental workflows.

Lin's Grand Buffet

Tucson, AZ

Expeditor

July 2022 - Present

- Coordinated kitchen operations to maintain efficiency in a high-volume environment
- Communicated across staff roles to resolve customer requests during peak service periods
- Supported fast-paced workflow execution under time-sensitive conditions

Vail Academy and High School

Tucson, AZ

Site Tech Coordinator Shadow

Sep 2021 - Nov 2021

- Assisted with troubleshooting hardware, software, and network issues
- Supported technical setup for school events and classroom systems
- Gained exposure to IT support workflows and infrastructure maintenance

PROJECTS

High-Speed Arduino Sorting Machine | https://github.com/LuckieMusngi/Arduino-Skittle-Sorter_Quannel-Method

- Built a high-throughput Arduino-based color sorting system (~4.44 items/sec), faster than all publicly documented skittle sorters.
- Designed embedded control logic integrating sensors, servos, and real-time actuation
- Developed mechanical system using custom CAD-designed funnel and rotary trapdoor mechanism
- Optimized timing and coordination between hardware and control loop for stable operation

Tech: Embedded C/C++, Arduino, OpenSCAD, Motors, Circuit Design, Real-Time Systems

Crime in Tucson Data Analysis | <https://github.com/LuckieMusngi/Crime-in-Tucson-Analysis>

- Led analysis of multi-source crime datasets to identify patterns and predict crime levels
- Built data preprocessing pipeline using Pandas and NumPy for cleaning and feature engineering
- Trained and evaluated classification models using Scikit-learn
- Visualized trends and model performance using Matplotlib

Tech: Python, Pandas, NumPy, Scikit-learn, Machine Learning, Data Cleaning

Complete Normalized SQL Database | <https://github.com/LuckieMusngi/Ski-Resort-Database>

- Designed normalized relational schema with constraints and entity relationships
- Implemented SQL-based system with CRUD operations via Java JDBC
- Built CLI interface for database interaction and management
- Applied E-R modeling and relational design principles

Tech: Oracle SQL, Java, JDBC, Database Design, ER Modeling, CLI

Bank Website Management System | <https://github.com/LuckieMusngi/BankWebsite>

- Built full-stack banking web application with multi-page user workflows
- Developed backend APIs using Node.js and Express.js
- Integrated MongoDB for persistent data storage and retrieval
- Implemented session-based user state management across pages

Tech: *Node.js, Express.js, MongoDB, JavaScript, HTML/CSS*

Air Pollution vs. Health RDB | Air vs. Health RDB

- Designed normalized PostgreSQL database integrating large-scale environmental and health datasets
- Built Python ETL pipelines to clean and structure 100K+ real-world records for analysis
- Loaded processed datasets into cloud-hosted PostgreSQL database using psycopg2 and bulk inserts
- Performed relational analysis to identify correlations between environmental and health metrics

Tech: *Python, Pandas, NumPy, PostgreSQL, SQL, psycopg2, Matplotlib, Seaborn, Aiven*

Real-Time Procedurally Generated Game (Unity / C#) | <https://github.com/LuckieMusngi/Unity-Roguelike>

- Developed a roguelike Unity game with procedural level generation for replayable runs
- Implemented enemy AI, projectile systems, and real-time gameplay mechanics
- Built modular object-oriented architecture using Unity MonoBehaviour system
- Designed runtime event sequencing using coroutines for game flow control

Tech: *Unity, C#, Procedural Generation, AI, Game Systems*